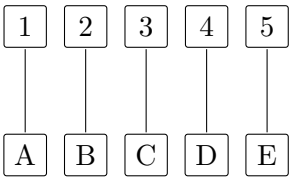
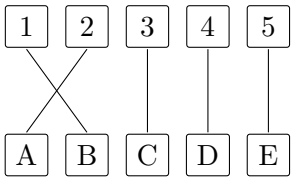
There are two classes of kids, and each class lines up. A complete match gives each kid a partner in the other class, with nobody left out. It keeps order when any two kids in the first class appear in the same order as their partners in the second class.

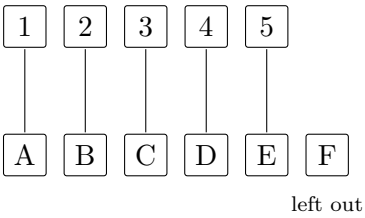
Complete match keeps order



Complete match does not keep order



5 kids and 6 no complete match possible

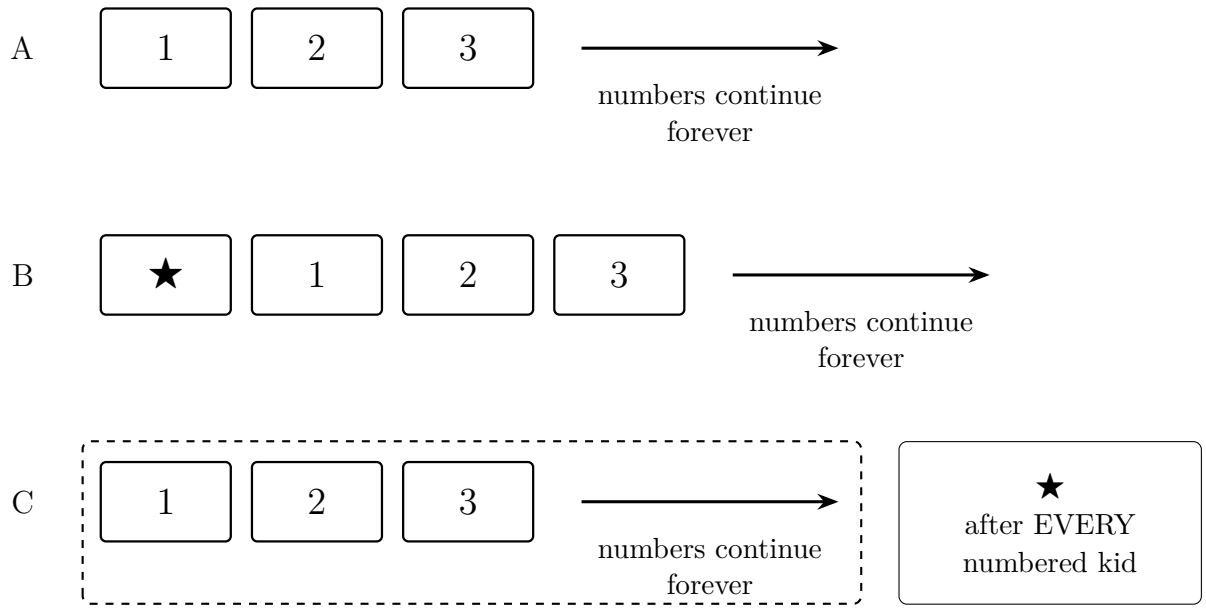


Problem 1: Find every complete match between these two class lines. Which matches keep order?

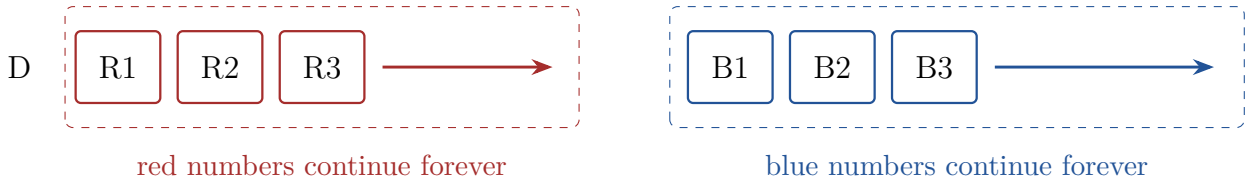
R1	R2	R3
B1	B2	B3

An arrow marked “numbers continue forever” represents all later numbered kids.

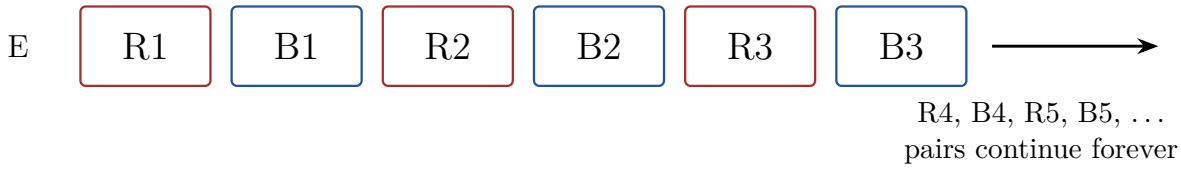
Problem 2: Labels only identify kids. Partners may have different labels; identical labels in two lines do not have to be paired. Compare A with B, then A with C. Can you match every kid? Can a complete match keep order? Give a rule for every kid, or explain why it is impossible.



Problem 3: Compare D with E. Can you match every kid? Can a complete match keep order? Give a rule for every kid, or explain why it is impossible.



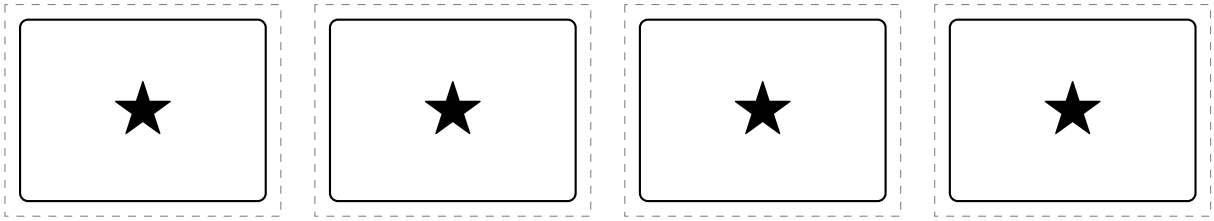
The whole blue row starts after EVERY red kid.



Problem 4: Arrange two endless numbered rows and up to two ★ kids on the table. A star or new row following a whole endless row comes after EVERY kid in that row.

Find two different arrangements that can be matched while keeping order. Find two that can be matched only if order may change. Sketch your arrangements in the boxes. Give matching rules or explain the obstruction.

R1	R2	R3	R4
R5	R6	R7	R8
R9	R10	R11	R12
B1	B2	B3	B4
B5	B6	B7	B8
B9	B10	B11	B12



This kid comes after EVERY kid
in the earlier row.

This kid comes after EVERY kid
in the earlier row.

This kid comes after EVERY kid
in the earlier row.

This whole row starts after EVERY kid
in the earlier row.

This whole row starts after EVERY kid
in the earlier row.